The Journal of Open Source Software

# Arcade: An urban building energy modeling tool featuring AI for input data enrichment

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**DOI:** 10.xxxxxx/draft

**Software**

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**Editor:**

**Submitted:** 18 February 2025
**Published:** unpublished

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## Summary

With rapid urbanization, more than half of the global population now lives in cities, a trend expected to continue in the coming decades (United Nations, 2015). Urban areas, while engines of economic growth and innovation, are also responsible for a significant share of global energy consumption and greenhouse gas emissions. Buildings, as key components of urban infrastructure, account for a substantial portion of this energy demand, driven by heating, cooling, and other operational needs (González-Torres et al., 2022).

In this context, Urban Building Energy Modeling (UBEM) has emerged as a powerful tool to analyze the energy use at the city scale (Salvalai et al., 2024). UBEM involves creating simulation-based models of building stocks, leveraging diverse datasets and computational tools to assess energy performance, evaluate retrofit strategies, and support sustainable urban planning.

`Arcade` is a free and open-source UBEM tool developed as an add-on extension for Blender, a 3D computer graphics software (Blender Foundation, 2025). `Arcade` incorporates a reduced-order building thermal model for efficient energy simulations and uses Blender's Python API to enable custom panels and user interfaces. Additionally, it features an AI-powered image recognition module to estimate building attributes often missing from geospatial datasets, such as building height and construction age. `Arcade` is designed for researchers, professionals, and students working in the fields of urban energy modeling and district thermal networks.

Key features of `Arcade` include:

- **3D Visualization**: Seamless integration with Blender for 3D building visualization and parameter configuration.
- **Energy Simulation**: Calculation of hourly heating and cooling demand using a simplified resistance-capacitance thermal model.
- **Automatic building data enrichment**: Gaps in building attributes from OpenStreetMap (OSM) datasets are automatically filled using either a user-defined probabilistic method or BRAILS, an AI-powered image recognition module (Cetiner et al., 2024).
- **Building archetypes**: Predefined archetypes act as proxies for real buildings, enhancing scalability and computational efficiency by representing variations in location, occupancy, and construction age.

## Statement of need

In recent years, several UBEM tools have been developed to support the analysis and optimization of energy demand in urban areas (Ali et al., 2021; Ferrando et al., 2020). Despite these advancements, one of the major challenges remains the availability of reliable geospatial datasets, including key building attributes such as geometry, construction year, and occupancy type, among others. These datasets are essential, as they are converted into input parameters

for the UBEM simulation engine after pre-processing and manipulation. Although governments worldwide are releasing more such datasets, these efforts are often fragmented (Biljecki et al., 2021). Therefore, researchers have increasingly relied on crowdsourced platforms such as OSM. These platforms provide free access to data and often offer better coverage than official sources. However, the quality and completeness can vary significantly between countries or even cities, thereby impacting the application of urban analytics (Wang et al., 2025). For instance, while the city of Berkeley has approximately 35,000 buildings mapped in OSM, only 1,500 include height-related information, and a handful of buildings have data about the construction year, which is particularly critical for categorizing buildings into archetypes for energy simulations.

In this context, `Arcade` differentiates itself from other UBEM tools by featuring a built-in functionality that leverages an AI-enabled module to automatically enrich incomplete OSM data using satellite and street-level imagery. The main simulation output consists of hourly data of heating and cooling demand of the urban area under investigation. This output data can be used for various analyses, including urban planning, retrofit scenario evaluation, and the design and optimization of district heating and cooling systems. The overall workflow of `Arcade` is illustrated in Figure 1.

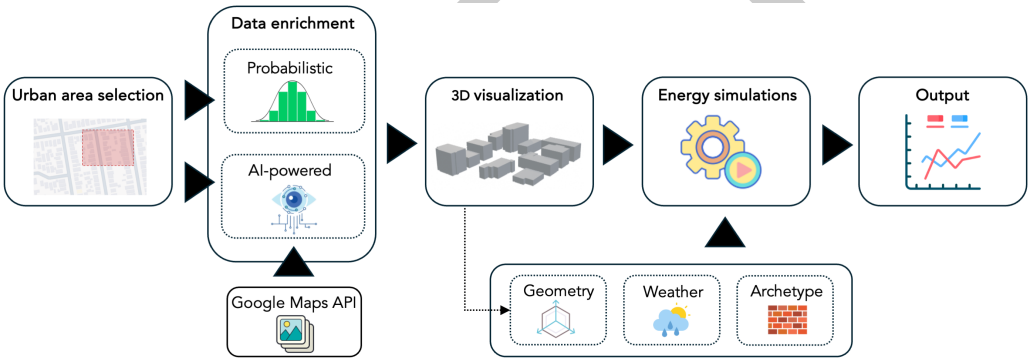


**Figure 1:** Arcade workflow.

# Acknowledgements

Support from Barbaros Cetiner in the use of BRAILS is gratefully acknowledged.

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